Action for EM field.

So far: particles on the <u>external</u> EM field are described by

$$S = S_{free} + S_{int} = -\sum_i m_{(i)} c^2 \int d\tau_{(i)} - \sum_i q_{(i)} \int d\tau_{(i)} A_\alpha u^\alpha_{(i)}$$

particle index

In this action, A_α is taken at the point along (i)-th particle world-line.

But now we would like to describe A_α itself as a dynamical field which is produced by particles.

$\Rightarrow$ this requires <u>field theory</u>: 4-potential $A_\alpha(t, \vec{r})$

at <u>every</u> point of Minkowski space serves as generalized coordinate. The 4-gradient $\frac{\partial A_\alpha(t,\vec{r})}{\partial x^\beta}$

is then generalized velocity.

Then Lagrangian $L(t) = c \int dV \, \mathcal{L}\left(A_\alpha(t,\vec{r}), \frac{\partial A_\alpha(t,\vec{r})}{\partial x^\beta}\right)$

spatial volume element

Lagrangian density

$$S_{field} = \int dt \, L = \int d\Omega \, \mathcal{L}\left(A_\alpha, \frac{\partial A_\alpha}{\partial x^\beta}\right)$$

$$d\Omega = c\,dt\,dV = c\,dx^0 dx^1 dx^2 dx^3$$

(do not confuse it with the spherical angle!)

$d\Omega' = d\Omega$ because Jacobian of Lorentz transformation $J = \gamma^2(1-\beta^2) = 1$.

Hence $\mathcal{L}$ has to be 4-scalar.

Demand also that $\mathcal{L}$ is quadratic in A_α $\Rightarrow$ in this case equations which we obtain by varying A_α will be linear $\Rightarrow$ and this is required for the <u>superposition</u> principle to hold.

Candidates for 4-scalar: $F_{\alpha\beta} F^{\alpha\beta}$ (!), $\cancel{\varepsilon_{\alpha\beta\mu\nu} F^{\alpha\beta} F^{\mu\nu}}$, $\cancel{A_\alpha A^\alpha}$

full derivative

Proca term. $\Leftarrow$ violates gauge invariance

$$\Rightarrow \quad S_{field} = \cancel{2}\, \mathcal{C} \int d\Omega \, F_{\alpha\beta} F^{\alpha\beta}$$

$$= 2\mathcal{C} \int d\Omega \left(\vec{B}^2 - \frac{1}{c^2} \vec{E}^2 \right)$$

Later arguments tell that $\mathcal{C} < 0$.

SI system of units: $\mathcal{C} = -\frac{1}{4\mu_0 c}$

μ_0 = magnetic permeability of vacuum.

$$S_{field} = -\frac{1}{4\mu_0 c} \int d\Omega \, F_{\alpha\beta} F^{\alpha\beta} = -\frac{1}{2\mu_0 c} \int d\Omega \left(\vec{B}^2 - \vec{E}^2/c^2 \right)$$

$$= -\frac{1}{2\mu_0 c} \int d\Omega \left\{ (\vec{\nabla} \times \vec{A})^2 - \frac{1}{c^2} \left(\frac{\partial \vec{A}}{\partial t} + \vec{\nabla} \varphi \right)^2 \right\}$$

How do we fix the sign of $\mathcal{C}$?

$$S_{field} = 2\mathcal{C} \int dt \left\{ (\vec{\nabla} \times \vec{A})^2 - \frac{1}{c^2}\left(\frac{\partial \vec{A}}{\partial t} + \vec{\nabla}\varphi\right)^2 \right\}$$

Let us <u>choose</u> $\vec{B}=0$ in some volume, but constant $\vec{E}_0$.

Gauge: $\varphi = 0$

$\Rightarrow \vec{A}(t) = -\vec{E}_0(t)$ is vector potential for this situation.

Consider now <u>variation</u> $\vec{A}(t,\vec{r}) = -\vec{E}_0 t(1+\eta(t))$

with $\eta(t_1) = \eta(t_2) = 0$.

$$\Rightarrow S_{field} = -2\mathcal{C}\frac{V}{c^2}\int_{t_1}^{t_2} dt \left(\frac{\partial}{\partial t}\vec{E}_0 t(1+\eta(t))\right)^2 =$$

$$= -2\mathcal{C}\frac{V}{c^2}\int dt \left(\vec{E}_0(1+\eta(t)) + \vec{E}_0 t\,\eta'(t)\right)^2 =$$

$$= -2\mathcal{C}\frac{V}{c^2}E_0^2 \int dt \left(1+\eta(t)+t\eta'(t)\right)^2$$

$$= -2\mathcal{C}\frac{VE_0^2}{c^2}\int_{t_1}^{t_2} dt \Big[\underset{\text{const}}{1} + 2\underbrace{(\eta + t\eta')}_{\frac{d}{dt}[t\eta(t)]} + \underbrace{(\eta+t\eta')^2}_{\geq 0}\Big]$$

full derivative

$\Rightarrow \int(\eta + t\eta')^2 dt$ dictates that $\mathcal{C}$ must be negative

so that S_{field} has minimum at $\eta(t)=0$.

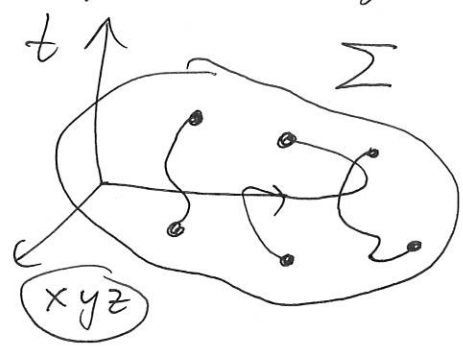
This fixes the sign: $\mathcal{C} < 0$.

$$\mathcal{C} = -\frac{1}{4\mu_0 c}.$$

One more point: the choice of $\int_{t_1}^{t_2} dt\, L$ is frame specific.

Even if there is a frame where trajectories of all particles begin at the same t_1, this will no be true in other reference frames. $\Rightarrow$ need to generalize.

Covariant formulation: the integral is over some 4-volume bounded by closed hypersurface Σ in Minkowski space. The field A^α is fixed everywhere on Σ but is free to vary inside it. Particle trajectories begin on Σ and end on Σ too.



Total action

$$S = S_{free} + S_{field} + S_{int}$$

must have minimum for the true motion of particles and true dynamics of EM fields.

Charge density and 4-current

$q_{(i)} \to \rho(t,\vec{r})$ charge density field

$dq = \rho(t,\vec{r})\, dV$ charge inside dV.

dV is small enough so that all charges inside move with the same velocity $\vec{v}(t,\vec{r})$

$\Rightarrow \vec{v}_{(i)} \to \vec{v}(t,\vec{r})$ is the velocity field

$$\sum_{(i)} q_{(i)}\, f(t_{(i)}, \vec{r}_{(i)}) \to \int dV\, \rho(t,\vec{r})\, f(t,\vec{r})$$

for any f.

Then $S_{int} = -\sum_{(i)} q_{(i)} \int dx^\alpha_{(i)}\, A_\alpha$

here $dx^\alpha_{(i)} = u^\alpha_{(i)}\, d\tau_{(i)} = u^\alpha_{(i)} \dfrac{dt}{\gamma_{(i)}}$

$$S_{int} = -\sum_{(i)} q_{(i)} \int dt\, \frac{u^\alpha_{(i)}}{\gamma_{(i)}} A_\alpha$$

$$\Rightarrow -\int dt\, dV\, A_\alpha \frac{\rho u^\alpha}{\gamma} = -\frac{1}{c}\int d\Omega\, A_\alpha j^\alpha$$

$d\Omega = dx^0 dx^1 dx^2 dx^3 = c\, dt\, dV$

$\Rightarrow$ 4-current $j^\alpha = \rho u^\alpha/\gamma = \rho(c, \vec{v}) = (\rho c, \vec{j})$

$u^\alpha = \gamma(c, \vec{v})$ $\qquad$ $\vec{j} = \rho\vec{v}$ current density

$$j_\alpha = (\rho c, -\vec{j})$$

ρ and $\vec{j} = \rho\vec{v}$ are connected by <u>charge conservation</u> which must be satisfied <u>locally</u> (in space-time).

$$\frac{\partial q}{\partial t} = \frac{\partial}{\partial t}\int dV\,\rho = -\oint_S d\vec{a}\cdot\vec{j} = -\int_V dV\,\vec{\nabla}\cdot\vec{j}$$

divergence theorem.

$$dq = \rho\vec{v}\cdot \underset{\substack{\uparrow \\ \text{surface area}}}{d\vec{a}}\,dt = \vec{j}\cdot d\vec{a}\,dt$$

The minus sign of $\oint d\vec{a}\cdot\vec{j}$: an outward (positive) flow of charge <u>decreases</u> the amount of charge inside dV.

$$\Rightarrow \frac{\partial\rho}{\partial t} = -\vec{\nabla}\cdot\vec{j} \Rightarrow \boxed{\frac{\partial j^\alpha}{\partial x^\alpha} = 0}$$ Lorentz invariant!

Gauge symmetry: $\mathcal{S}$ physical fields do not change if $A_\alpha \to A'_\alpha - \frac{\partial g}{\partial x^\alpha}$ for some arbitrary $g(t,\vec{r})$.

But $S_{int} = \frac{1}{c}\int d\Omega\, A_\alpha j^\alpha \to S'_{int} = \frac{1}{c}\int d\Omega\, A'_\alpha j^\alpha$

$$-\frac{1}{c}\int d\Omega\,\frac{\partial g}{\partial x^\alpha} j^\alpha$$

$$\frac{1}{c}\int d\Omega\,\frac{\partial g}{\partial x^\alpha} j^\alpha = \underbrace{\frac{1}{c}\int d\Omega\,\frac{\partial(g j^\alpha)}{\partial x^\alpha}}_{\frac{1}{c}\int_\Sigma d\Sigma_\alpha (g j^\alpha) = \text{const}} - \frac{1}{c}\int d\Omega\, g\,\frac{\partial j^\alpha}{\partial x^\alpha}$$

since all fields $\mathcal{S}$ (including $g(t,\vec{r})$!) and charges are fixed on Σ.

$$\Rightarrow S'_{int} = \frac{1}{c}\int d\Omega\, A'_\alpha j^\alpha + \text{const} - \underbrace{\frac{1}{c}\int d\Omega\, g\, \frac{\partial j^\alpha}{\partial x^\alpha}}_{\phi}$$

because $\frac{\partial j^\alpha}{\partial x^\alpha} = 0$.

$\Rightarrow$ gauge invariance is a consequence of the physical <u>charge conservation</u> requirement.

Point charge: $\rho(t,\vec{r}) = q\,\delta(\vec{r}-\vec{r}_0)$

q at $\vec{r}_0(t)$

moving

with $\vec{v}_0(t) = \frac{d\vec{r}_0(t)}{dt}$

$$\vec{j}(t,\vec{r}) = q\,\vec{v}_0(t)\,\delta(\vec{r}-\vec{r}_0)$$

Indeed $\frac{\partial\rho}{\partial t} = q\frac{\partial}{\partial t}\delta(\vec{r}-\vec{r}_0(t)) = q\left(-\frac{\partial}{\partial\vec{r}}\delta(\vec{r}-\vec{r}_0)\right)\cdot$

$$\cdot\frac{d\vec{r}_0}{dt} = -q\,\vec{v}_0(t)\,\vec{\nabla}\delta(\vec{r}-\vec{r}_0(t)).$$

$$= -\vec{\nabla}\cdot\vec{j} = -q\,\vec{v}_0(t)\,\vec{\nabla}\cdot\delta(\vec{r}-\vec{r}_0(t)).$$